

## Lecture 4 Quiz Questions and Answers

1

The output of the feedback control system must be a function of

reference input only

☐ 8%

reference output only

☐ 3%

output and feedback signal

☐ 13%

input and feedback signal ☒

☐

none of the above

☐ 3%

2

Select the correct statement

characteristic equation simplifies the evaluation of a system's response

☐ 0%

characteristic equation are fundamental to the analysis and design of control systems

☐ 10%

characteristic equation can be obtained using zeros of the system

☐ 2%

1 and 2 ☒

☐

all of the above

☐ 23%

3

Select the correct statement, if a factor of the denominator can be cancelled by the same factor in the numerator

system should have equal number of poles and zeros

☐ 9%

all of the roots of the characteristic equation should be same as the zeros

☐ 0%

the root of this factor no longer causes the transfer function to become zero ☒

☒

1 and 2

☐ 20%

2 and 3

☐ 34%

4

Select the correct statement

system is stable if its transient response decays

☐ 9%

system is stable, if all the zeros are in the right hand s-plane,

☐ 7%

when system marginally stable, at least one pole is on the imaginary axis

☐ 20%

1 and 3 ☒

☒

all of the above

☐ 2%

5

If the response is a decaying sinusoid:

all the poles are on the left hand s-plane



both poles and zeros are on the left hand s-plane



complex conjugate pole pair on the left hand s-plane ✓



complex conjugate pole pair on the right hand s-plane



imaginary pole pair on the imaginary axis

